

Statement of Purpose

During my Bachelor's degree in Computer Science at Iqra University, I gradually realized that I was most interested in the part of computer science where software could learn from data and make sense of the world around us. My first exposure to Artificial Intelligence and Machine Learning made me curious about how computers could recognize patterns, learn from examples, and solve problems that normally require human intelligence. This curiosity became stronger when I started working with images and deep learning models. I became particularly interested in Computer Vision because it combines programming and problem-solving with the ability to help machines understand visual information. Over time, this interest developed into a clear academic goal: I want to deepen my knowledge of Artificial Intelligence, Machine Learning, and Computer Vision and learn how to develop intelligent systems that can be useful in real-world situations. One experience that had a major influence on this goal was my final-year project, AI-based Skin Lesion Detection and Classification. Working on this project allowed me to apply the concepts I had learned during my degree to a problem with real-world importance. I worked with the publicly available HAM10000 dermoscopic image dataset and developed a Convolutional Neural Network for skin lesion classification. My reported classification accuracy was 97.86%. While achieving this result was an important part of the project, the experience taught me much more than how to obtain a high accuracy score. It showed me how much work is involved in developing a reliable machine learning system, from preparing data and selecting a model to testing results and understanding the limitations of the approach. During the project, I studied different deep learning architectures, including ResNet, VGGNet, DenseNet, and EfficientNet, and explored concepts such as transfer learning and hybrid deep learning approaches. I also looked at challenges that are common in medical image analysis, including class imbalance, segmentation, feature extraction, and model generalization. To understand the broader computer vision pipeline, I studied approaches such as U-Net, Fast R-CNN, and YOLO for segmentation and detection. I also compared deep learning approaches with traditional machine learning methods such as Support Vector Machines, Random Forest, and K-Nearest Neighbors. What I found most valuable was learning that building an AI system is not simply about choosing an algorithm and training it. I had to understand the data, make decisions about preprocessing, experiment with different approaches, evaluate the results, and think critically about whether the model could work reliably outside the data used during training. This changed the way I think about Artificial Intelligence. I became more interested in the reasoning behind models and in the research questions that arise when AI is applied to real problems. In particular, working with medical images made me appreciate the importance of reliability and generalization when AI is used in areas where incorrect predictions can have serious consequences.

My Bachelor's degree also gave me a broad foundation in Computer Science. I studied programming, data structures, algorithms, databases, computer networks, and other core areas that

helped me develop the problem-solving skills needed for advanced study. At the same time, my interest gradually moved toward Artificial Intelligence and Computer Vision. Courses and practical projects introduced me to areas such as image classification, feature extraction, machine learning, and deep learning. Instead of treating these subjects only as academic requirements, I tried to explore them through practical implementation. This approach helped me understand concepts more clearly and also made me more confident in working independently on technical problems.

Alongside my academic and AI experience, I have developed practical experience as a software developer. I believe this background complements my interest in Artificial Intelligence because an AI model is most useful when it can be integrated into a working system. During my internship as a Full Stack Developer at Interarch Technologies, I worked on responsive web applications and RESTful APIs using React.js, Next.js, Node.js, and MongoDB. I also contributed to frontend optimization that resulted in a reported 30% reduction in page load times. This experience taught me how to think about software from a practical perspective, including performance, usability, data handling, and communication between different parts of an application.

At Code Alpha, I worked on frontend development and data-related tasks, including creating reusable React components and integrating applications with RESTful APIs. These experiences helped me improve my understanding of how real applications are developed and maintained. I also worked independently on projects such as an online shopping platform using Java, MySQL, and JDBC and a web-based bus ticket booking system using PHP and MySQL. Through these projects, I gained practical experience with databases, application logic, system design, and developing software according to user requirements.

My technical experience includes Python, OpenCV, TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, and Pandas for machine learning and computer vision work. I have also worked with modern web technologies including React, Next.js, Node.js, and MongoDB. Having experience in both software engineering and Artificial Intelligence has helped me see the connection between developing a model and building a complete application around it. I now want to strengthen this combination through graduate-level study and learn how advanced AI techniques can be developed and applied in a more systematic way.

I have chosen Germany because I want to continue my education in an environment where theoretical knowledge and practical research are closely connected. Germany has a strong academic and research culture in computer science and Artificial Intelligence, and I am particularly

attracted to the opportunity to learn in an international environment while working with researchers and students who have different perspectives and experiences. I am interested in research environments such as BIFOLD at TU Berlin, the Heinz Nixdorf Institute at Paderborn University, and the Lamarr Institute for Machine Learning and Artificial Intelligence associated with the University of Bonn. Their work in areas such as machine learning, intelligent systems, data-driven research, and applied AI is closely related to the direction in which I want to develop.

What I am looking for in a Master's degree is not simply another qualification. I want to understand Artificial Intelligence at a deeper level. During my undergraduate project, I often found myself asking questions that could not be answered simply by following a tutorial or using an existing model. Why does one architecture perform better than another? How can models become more robust when datasets are limited or imbalanced? How can computer vision systems be made more reliable when they are applied to real-world data? How can an AI model be integrated into a system in a way that makes it genuinely useful? These questions have motivated me to pursue further study.

A Master's program will give me the opportunity to strengthen my mathematical and theoretical understanding of machine learning while developing more advanced research and practical skills. I want to become comfortable not only with applying existing techniques but also with understanding their limitations, designing experiments, interpreting results, and exploring new approaches. I am especially interested in developing further expertise in deep learning and computer vision, with the possibility of applying these skills to healthcare and other areas where visual information can help solve practical problems.

My long-term goal is to build a career in Artificial Intelligence and Computer Vision, either in an industrial research and development environment, an AI-focused organization, or an academic research setting. I would like to work on systems that solve genuine problems rather than developing models only for their technical performance. My experience with skin lesion classification has particularly influenced my interest in applications where AI can assist people and provide meaningful value. At the same time, I want to remain open to other areas of Computer Vision as I gain more knowledge and research experience during my Master's degree.

I am also open to pursuing a PhD in the future if my Master's research leads me toward a research problem that I want to explore further. For now, my immediate goal is to become a stronger AI researcher and engineer by building a deeper understanding of machine learning, deep learning, and computer vision and by learning how to conduct research in a rigorous academic environment.

Looking back at my undergraduate journey, my interest in AI has grown through a combination of coursework, experimentation, research, and software development. My final-year project showed me the challenges and possibilities of applying deep learning to real-world images, while my software development experience taught me how to turn technical ideas into functional systems. These experiences have given me a clear direction for my next step.

I believe that studying Artificial Intelligence, Machine Learning, and Computer Vision at the Master's level in Germany will allow me to build on what I have already learned while challenging me to think more deeply and independently. I am ready to learn from experienced researchers, work with other students, take on challenging problems, and contribute my own experience in software development and computer vision. Most importantly, I want to continue developing the curiosity that first attracted me to Artificial Intelligence and turn it into the knowledge and research skills needed to create intelligent systems that are reliable, practical, and genuinely useful.